

SHUHUI ZHAO

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SKILLS

- Statistical and mathematical modelling
- Performing data analysis using Excel, Power BI, MySQL, R, and Python
- Constructing API in Node.js and Python
- JavaScript, React, Svelte, TypeScript, HTML, CSS, Git

PERSONAL PROJECTS

- Image Generator:
 - Created a full-stack application for generating images using **OpenAI** and creating blog posts
 - Used **React** and **React Router** to create web applications with well-defined routes and UI components
 - Used **style-components** to apply **custom CSS** with isolated scopes
 - Used **Cloudinary** and **MongoDB** to store generated images and blog post contents
 - Used **Express.js** along with **Mongoose** and **OpenAI SDK** to interact with **MongoDB** and **OpenAI**
- Rich-text Editor:
 - Created a Google Doc clone that allows collaborative rich-text editing.
 - Used **Socket.io** to stream data changes between users within the same collaborating editing session
 - Used a **Node.js** server and **Mongoose** to handle the communication to **MongoDB**
 - Used **React** and **Quill** to create a UI for live editing and document change streaming
- Vocabulary App:
 - Create a vocabulary app for memorizing and testing French vocabulary
 - Used **Svelte** with **Typescript** to create UI components
 - Created **Typescript** type definition to strictly define the contract between frontend and backend
 - Used **json-server** as the backend to store data
- Handwritten Digit Recognition with Neural Networks (MNIST dataset)
 - Implemented the mini-batch gradient descent algorithm from scratch
 - Constructed and trained large neural networks using **R** libraries including **Keras** and **r-Tensorflow**

PROFESSIONAL EXPERIENCES

MNP LLP (Ottawa)

Consultant, Enterprise Risk – Technology Risk

2023.10 – present

Analyst, Enterprise Risk – Technology Risk

2022.09 – 2023.09

- IT audits and Computer-Assisted Audit Techniques (CAATs)
 - Assisted clients in data extraction, enabling efficient retrieval of financial data from **SQL databases** for audit and reconciliation
 - Developed **Python** scripts using **Numpy** and **Pandas** to reperform calculations, including opening balance, closing balance, and expected revenue, by analyzing historical data
- System transformation engagements
 - Led the development of a **Power BI** model to reconcile clients' general ledger account balances during system conversion and produce detailed reports
- Model validation
 - Validated various financial reporting models including CRA (Credit Risk Assessment) superficial loss model using **Excel VBA**

EDUCATION

Carleton University, Ottawa Canada

Master of Science, Statistics (11.875/12)

2020.09 – 2022.02

Carleton University, Ottawa, Canada

Qualifying-Year Program, Biostatistics

2019.09 – 2020.05

Capital Normal University, Beijing, China

Bachelor of Science, Biology (3.96/5, top 5%)

2013.09 – 2017.06

University of Tübingen, Tübingen, Germany

Exchange Student (Conducted experiments for undergraduate research & thesis)

2016.10 – 2017.03